

Can Labor Market Opportunities Explain the Rising Levels of Morbidity and Mortality in the USA?*

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August 17, 2025

Abstract

I study the impact of large plant openings on mortality rates associated with Drug Overdose, Suicide, and Alcoholic Liver Disease, collectively termed Deaths of Despair. Using a natural experiment proposed by [Greenstone and Moretti \(2003\)](#), I exploit the exogenous opening of Million Dollar Plants (MDPs) across US counties between 1982 and 1993 to causally estimate the effects of improved labor market conditions on mortality. My findings reveal that counties attracting large plants experienced significant reductions in overdose mortality of approximately 2.60 deaths per 100,000 population per year, a 23.3 percent decrease relative to the mean overdose rate, while suicide and alcoholic liver disease mortality were unaffected. The effects persist and intensify over time, and are strongest among men, midlife adults, and White individuals. Consistent with a labor-market channel, winning a plant raised county employment by roughly 10 percent and total wages by 9-10 percent, while placebo tests on unrelated causes of death rule out a general improvement in health. The results are robust to alternative specifications and to a range of checks confirming that they are not driven by the differential regional timing of the opioid epidemic, diverging trends in opioid supply, or state-level opioid regulations.

*I am indebted to Samuel Bazzi for generous advice and support on this project. I also want to express my gratitude with Jan Stuhler and Samuel Bentolila for their guidance. I thank Thomas Rieger, Aurel Rochell, Daniel Sturm, and participants at the 2nd Berlin PhD Conference for helpful comments. Errors are my own.

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1 Introduction

In the aftermath of the Second World War, America experienced what some labeled the “*Golden Age of Capitalism*” (Marglin and Schor, 1990). The period between 1945 and 1973 saw a steady and steep improvement in living standards among the US population, along with unprecedented labor-productivity growth averaging 2.8 percent per year in the nonfarm business sector (U.S. Bureau of Labor Statistics, 2011). Rising levels of demand were matched with advances in technology and increasing productivity, allowing for simultaneous improvements in wages and employment. Although economic growth continues to be a feature of post-war America up to today, its pace slowed down after the 1970s. The virtuous dynamics of the Golden Era were slowly reversed by a structural change that transformed the US from an industrialized to a service-based economy, with less room for increases in productivity and wages (Iversen and Wren, 1998; Iversen and Cusack, 2000).

The picture of America that we see today is rather at odds with tales of post-war prosperity. Increasing economic integration and robotization have challenged the structure of US labor markets in the past decades, particularly for low- and middle-skilled workers, who have been disproportionately hurt by the growth in occupational upgrading and technological change (Brynjolfsson and McAfee, 2014; Autor and Dorn, 2013; Goldin and Katz, 2007; Acemoglu and Restrepo, 2020). A further sign of American decline was uncovered by Case and Deaton (2015), who documented rising levels of morbidity and early mortality among middle-aged adults. Beginning in the 1990s, they observed an increase in deaths caused by Drug Overdose, Suicide, and Alcoholic Liver Disease. Due to the morbid nature of these fatalities, they coined the term “*Deaths of Despair*”. Figure A.1 illustrates the evolution of Deaths of Despair per 100,000 population, disaggregated by category from 1975 to 2022. Panel A.1a provides unweighted crude rates, while Panel A.1b weights the crude rates by the population of each county. Weighted rates are comparable to unweighted ones but generally somewhat larger, suggesting that more populated counties experienced greater mortality; the main driver of the increase in deaths is drug overdoses. Figure A.2 in the Appendix breaks down the data by gender, age, and race and shows that the burden of the deaths has been disproportionately borne by male adults aged 35 to 54 years old. Finally, Figure A.3 presents state-level maps comparing the prevalence of Deaths of Despair per 100,000 population in 1970 and 2022.¹ Although the prevalence of Deaths of Despair has more than doubled in most states, the Appalachian, Mountain West, and New England regions have experienced the biggest burden.

The goal of this paper is to analyze whether improved labor market opportunities at the county level can mitigate or reverse the rising prevalence of Deaths of Despair. A key challenge in addressing this question lies

¹Displaying county-level aggregates was forbidden by the Centers for Disease Control and Prevention (CDC) in 1989. Therefore, I only show state-level aggregates. In my analysis, I use county-level aggregates.

in isolating the causal effect of labor market conditions on social outcomes from other confounding factors. To overcome this challenge, I leverage a natural experiment first exploited by [Greenstone and Moretti \(2003\)](#). Between the 1980s and early 1990s, local governments in the US competed with each other for large industrial plants to settle within their jurisdictions. I examine the impact of successfully attracting such plants on the rates of Deaths of Despair. By exploiting the arguably exogenous variation in labor market conditions created by the openings of these large plants, I estimate the causal effects of improved employment opportunities. I implement a staggered differences-in-differences estimation to compare winner counties that attracted the plants to those that were the second-best option and barely lost the competition. Several robustness checks are also conducted to verify the validity of the baseline estimations.

Results suggest that each county that successfully attracted a large plant experienced an average annual reduction of between 2.49 and 2.60 overdose deaths per 100,000 population, representing a 22.3-23.3% decline relative to the sample mean (Table [A.4](#)). These reductions should be read against the size of the labor-market shock that produced them: winning a plant raised county employment by about 10 percent, the number of establishments by roughly 8 percent, and total annual wages by 9-10 percent (Table [A.7](#)). The reduction was most pronounced among middle-aged adults and middle-aged men. Moreover, the effect persisted and intensified over time. These findings are robust to various alternative specifications. Robustness checks include testing for diverging trends in opioid supply between winning and losing counties; controlling for variations in state regulations governing legal substances (e.g. Oxycodone, Fentanyl, Codeine, Hydrocodone, and Morphine); and conducting placebo tests on diseases unrelated to despair but associated with general health conditions. These checks consistently support the validity and strength of the baseline results.

To gauge the magnitude of these effects, it is useful to place them both against the labor-market shock that produced them and against comparable estimates in the literature. A roughly 10 percent increase in county employment lowers overdose mortality by 2.60 deaths per 100,000, so that each percentage point of employment growth is associated with about 0.27 fewer overdose deaths per 100,000, an implied elasticity of overdose mortality with respect to employment of approximately -2.4 .² This magnitude is economically meaningful but not implausibly large when set beside the demand-side estimates it mirrors. [Venkataramani et al. \(2020\)](#) find that automotive plant closures raised opioid mortality by 8.6 deaths per 100,000, and [Pierce and Schott \(2020\)](#) attribute up to 3 additional overdose deaths per 100,000 to Chinese import competition. The plant openings studied here move mortality in the opposite direction and by a comparable order of magnitude. It is notable, however, that the reduction produced by a positive employment shock (2.60) is smaller than the increase [Venkataramani et al. \(2020\)](#) document from a negative one (8.6). This asymmetry

²This figure should be read as the effect of the broad labor-market improvement summarized by its employment component, since plant openings also raise wages, establishments, and local income, rather than as the effect of headcount alone.

is consistent with the loss-aversion logic discussed below: the mortality cost of destroying good jobs appears to exceed the mortality benefit of creating them (Tversky and Kahneman, 1991).

This article is structured as follows. Section 2 surveys the relevant literature. Section 3 describes the setting and the natural experiment. Section 4 describes the data sources used in the analysis. Section 5 details the methodology employed to estimate the causal effects (e.g. staggered differences in differences methods). Section 6 presents the baseline results. Section 7 discusses various robustness checks conducted to assess the reliability of the results. Section 8 concludes.

2 Literature Review

Case and Deaton (2015) were the first to draw attention to the rising levels of morbidity and early mortality in the United States. They observed a stagnation in the life expectancy of some demographic subgroups. Importantly, they also documented a rise in what they coined Deaths of Despair, an umbrella term that refers to three types of fatalities: deaths associated with drug overdose, alcoholic liver disease, and suicide. They noted two regularities in this phenomenon. First, Deaths of Despair were an American idiosyncrasy, and other comparably developed nations were not experiencing increases in morbidity or early mortality. Second, the rising mortality rates were concentrated among middle-aged White Non-Hispanic Americans. However, Case and Deaton (2015) simply documented this phenomenon, remaining silent on its drivers. Since then, the literature on the causes of Deaths of Despair has evolved, with an emphasis on deaths associated with drug overdose. Broadly, two contrasting strands of the literature have emerged. On the one hand, some authors advance a demand-side argument, and claim that Deaths of Despair are driven by deteriorating labor market conditions. On the other hand, some scholars underscore a supply-side argument, and claim that these fatalities are driven by easier access to opioid prescriptions.

On the demand side, Case and Deaton (2017) proposed the *Cumulative Disadvantage* hypothesis, which attributes these trends to the deterioration of social and labor market conditions among less-educated workers. They emphasized the erosion of “*the good jobs, those that generated a sense of identity and belonging among workers,*” which was compounded by an expensive healthcare system and the “*destruction of a way of life,*” rooted in values like family, community, and religion. In line with this hypothesis, Dow et al. (2019) find that states with higher minimum wages report fewer non-drug suicides, estimating that a 10% increase in the minimum wage reduces non-drug suicides among less-educated adults by 2.7%. Using a differences-in-differences design, Venkataramani et al. (2020) examine the impact of automotive plant closures in the industrial Midwest. They find that closures between 1999 and 2016 led to an 85% increase in opioid mortality rates, equivalent to 8.6 excess deaths per 100,000 inhabitants. Similarly, O’Brien et al. (2022), using

a shift-share instrument, report that automation accounted for 12% of the rise in drug overdose mortality between 1993 and 2007.

An important strand of the demand-side literature connects the “*China Shock*”³ to rising mortality rates. [Pierce and Schott \(2020\)](#) estimate that higher exposure to import competition led to up to 3 additional drug overdose deaths per 100,000 people. [Autor et al. \(2019a\)](#) examine variations in exposure at the commuting-zone level, and highlight declines in marriage rates, increased single parenthood, higher rates of child poverty, and elevated mortality from drug and alcohol abuse in more exposed areas. Finally, aligning with Case and Deaton’s emphasis on the role of social structures in shaping Deaths of Despair, [Giles et al. \(2026\)](#) highlight the critical influence of religiosity. Using the repeal of the “*Blue Laws*”⁴ as an exogenous shock to religiosity, they show that these deaths were often preceded by a decline in religious participation.

Diverging from demand-driven explanations, a second group of scholars emphasize a supply-side mechanism, grounded in the idea that overdose deaths are largely driven by the misuse⁵ of prescription drugs and narcotics. These authors highlight the unique characteristics of the U.S. drug landscape, which underwent major changes in pain management beginning in the 1990s. In December 1995, the FDA approved Purdue Pharma’s OxyContin, a potent medication intended to relieve moderate to severe pain. Purdue’s aggressive marketing strategy, mirrored by other pharmaceutical companies, spurred a dramatic rise in opioid prescriptions. Indeed, this widespread misuse of prescription opioids culminated in Purdue Pharma accepting responsibility in a landmark settlement.

[Alpert et al. \(2022\)](#) find that the increased availability of opioids following the introduction of OxyContin in 1995 explains a significant share of overdose deaths across counties. Similarly, [Cutler and Glaeser \(2021\)](#) demonstrate that the epidemic was more severe in areas with higher preexisting levels of pain and opioid shipments. In this spirit, [Ruhm \(2018\)](#) analyzes local measures of unemployment, income, prices, and import competition, and finds that economic conditions explain at most 10% of the observed rise in drug deaths since the turn of the century, while opioid prescriptions alone account for approximately 85% of this increase. Nonetheless, it is important to recognize that the latter studies focus exclusively on deaths related to drug overdose. As a result, these critiques do not address the rise in deaths associated with suicide or alcoholic liver disease.

This paper addresses a critical gap in the literature by exploring whether exogenous improvements in labor market prospects reduce mortality. By analyzing how greater labor market opportunities generated by

³Greater exposure to Chinese imports has been consistently linked to persistent declines in employment, labor force participation, and earnings ([Autor et al., 2013, 2019b](#); [Pierce and Schott, 2020](#)).

⁴Blue laws restricted Sunday activities (e.g. shopping, liquor sales, and recreational activities), and varied by state and locality across the USA.

⁵Misuse is defined as opioid use that deviates from the prescribed pattern, regardless of whether harm occurs. This includes taking higher doses than prescribed, using the medication more frequently, or acquisition through unauthorized routes.

the exogenous opening of large plants across the U.S. impacted the evolution of Deaths of Despair in counties attracting these plants, I implicitly test the validity of the Cumulative Disadvantage hypothesis. To the best of my knowledge, this paper provides the first direct causal evidence of the effect of improvements in labor market conditions on Deaths of Despair. Notably, existing research has focused exclusively on the effects of job loss on morbidity and mortality. This distinction is important because the effects of unemployment may not mirror the impacts of employment, either in magnitude or in nature. A central concept in behavioral economics is the notion of “*Loss Aversion*” (Tversky and Kahneman, 1991), the idea that people weigh losses more heavily than equivalent gains. As such, the negative consequences of massive job losses in a county may outweigh the perceived benefits of large plant openings.

3 Setting

I leverage a natural experiment first exploited by Greenstone and Moretti (2003), which captures the competitive process behind the location decisions of large plants in the United States. Between 1982 and 1993, several counties in the United States competed with each other in order to attract large industrial plants. The competition processes were covered by several issues of the corporate magazine *Site Selection*. This magazine recurrently featured a section titled “*Million Dollar Plants*” (MDPs) that provided in-depth analysis of how firms selected locations for major facilities. These firms typically began with a broad list of potential sites, which they systematically narrowed down to approximately ten contenders. The final stage involved two or three counties that were closely evaluated as finalists based on their potential to maximize profits. This rigorous selection process ultimately identified a “winner” county, where the plant was established, along with one or two “runner-up” counties that were seriously considered but eventually not chosen. Each MDP article detailed the chosen location (winner) and the runner-up counties (losers) that narrowly missed being selected in the last stage of the selection process. These articles not only identified the counties involved in the selection process, but also provided detailed information about the firms, the industry sectors in which they operated, and the year of announcement for each large plant’s decision.

Table A.1 in the Appendix reports descriptive statistics on the number of firms by sector, year of plant opening, and number of winner and loser counties. The first column reports the original sample, while the second restricts attention to the sample selected for this paper, which I detail in this section. Plant location announcements took place between 1982 and 1993. During this period, a total of 211 county-case observations reached the final stage of the selection process. Of these, 82 ultimately won the bid and hosted the large plant, while 129 lost. There are thus 82 bidding processes (or “cases”), each featuring a unique winner and at least one corresponding loser. Most of the plants operated in manufacturing (76.8%), followed

by transportation and utilities (9.8%), services (7.3%), trade (4.9%), and financials (1.2%). Nevertheless, I apply some trimming to this sample. First, 14 counties were losers in earlier cases but became winners in subsequent ones; because a county that is treated in one competition cannot serve as a clean control in another, their loser (control) observations are dropped as contaminated controls. Table A.2 in the Appendix lists these 14 counties and the competitions each won and lost. Second, two counties, Santa Clara and San Mateo (California), are dropped because they are the home bases of the competing firms (Hewlett-Packard and Alumax, respectively), and thus do not constitute clean counterfactuals. Finally, I drop the Toyota case (whose winning county already appears as a winner in an earlier Toyota competition, and whose sole losing county, Alameda, California, went on to host the NUMMI plant), together with Marion County (Indiana), which appears as both a winner and a loser across two United Airlines competitions. After these exclusions, the final sample comprises 80 cases and 188 county-case observations (80 winners and 108 losers). These exclusions leave the sample composition largely unchanged (Table A.1). Figure A.4 shows the geographic distribution of winner and loser counties across the continental United States.

My research design relies on a key identifying assumption: that runner-up locations, having undergone the same rigorous selection process and narrowly missed being chosen, serve as credible counterfactuals for the winners. Table A.3 compares winner and loser counties in the years before the plant openings and finds no statistically significant differences in pre-existing economic or demographic characteristics, supporting the use of these openings as a source of exogenous variation in local labor-market conditions. This setting thus provides a natural experiment for examining how exogenous improvements in labor markets shape the trajectory of Deaths of Despair.

4 Data

4.1 Mortality Data

I use proprietary *detailed all-county mortality files* obtained through petition to the National Center for Health Statistics (NCHS). These files include the complete universe of mortality records filed in the U.S. between 1970 and 2022. The records provide detailed information on the cause and location of death.⁶ Causes of death are classified by the NCHS using the International Classification of Diseases (ICD) codes. I employ ICD-10 codes for 1999 to 2022, ICD-9 codes for 1979 to 1998, and ICD-8 codes for 1970 to 1978. Details on the conversion between these versions can be found in the [Data Appendix](#). Access to confidential detailed mortality data also includes information on the race, gender, age, and educational attainment of each

⁶Descriptive statistics for the dependent variables cannot be displayed due to the confidentiality agreement. Specifically, the release of county-level mortality data was restricted after 1989 by the CDC.

individual, which I use to construct county-level yearly counts of total Deaths of Despair, as well as counts disaggregated by cause and demographic subgroups. I construct mortality counts by two gender subgroups (male and female), three race subgroups (White, Black, and “other” races),⁷ and three age subgroups: young (15 to 34 years old), midlife (35 to 54 years old), and old (55 years and older).

4.2 Population Data

I use population data from the Surveillance, Epidemiology, and End Results Program (SEER) of the National Cancer Institute (NCI), which provides detailed population estimates disaggregated by race, gender, and age group. These population estimates are matched to yearly death counts at various levels of disaggregation. I compute “crude” mortality rates, expressed per 100,000 population. The crude death rate for a county-year is calculated as the total number of deaths in the county-year divided by its total population in that year. County population also serves as the weight in the estimation.

4.3 Employment Data

I use county-level labor-market data from the Quarterly Census of Employment and Wages (QCEW), compiled by the Bureau of Labor Statistics (BLS). The QCEW provides an annual census of establishments, employment, and wages covered by unemployment insurance. I use the annual county files from 1975 to 2022, harmonizing the older industry-classification format (through 2000) with the more recent format (from 2001 onward) and retaining the all-industry, all-ownership totals for each county-year. From these files I construct three measures of local labor-market activity, namely the number of establishments, average annual employment, and total annual wages, which I use both to characterize economic conditions in winner and loser counties and as outcomes when examining the labor-market effects of plant openings.

4.4 ARCOS Data on Opioid Shipments

I use data on legal opioid shipments from the Drug Enforcement Administration’s Automation of Reports and Consolidated Orders System (ARCOS) for the period 1997-2017. These data detail shipments by product and three-digit ZIP code and are publicly available through PDF reports, which are processed to construct the dataset. Following the methodology outlined by [Cutler and Glaeser \(2021\)](#), I aggregate the ARCOS data to the county level using a ZIP-to-county crosswalk provided by the U.S. Census Bureau ([U.S. Census Bureau, 2010](#)). The dataset includes information on aggregate opioid shipments, expressed in morphine

⁷Analysis of race categories beyond Black and White is limited due to the nature of population data from the Surveillance, Epidemiology, and End Results Program (SEER) of the National Cancer Institute (NCI). Before 1990, race categories were recorded as White, Black, or “other.”

milligram equivalents (MMEs), from 1997 through 2017. Specifically, it contains shipment data for five types of opioids: Oxycodone, Fentanyl, Codeine, Hydrocodone, and Morphine.

4.5 Data on State Opioid Laws

I incorporate data on state-level opioid legislation. Using state statutes, regulations, and existing surveys, [Meara et al. \(2016\)](#) constructed a dataset documenting legal restrictions on controlled substances for all states and the District of Columbia from 2006 to 2012. This dataset, freely available online, includes information on laws regulating patients, prescribers, and dispensing pharmacists. Specifically, the dataset captures legislation involving eight categories: quantitative prescription limits, patient identification requirements, physician examination requirements, pharmacist verification requirements, restrictions on doctor shopping, prescription drug monitoring programs (PDMPs), tamper-resistant prescription forms, and pain-clinic regulations. [Meara et al. \(2016\)](#) quantified state controlled-substance laws by indicating whether each of these eight types of laws was in effect throughout a given year in each state.

5 Methodology

I exploit the winner-runner-up research design of [Greenstone and Moretti \(2003\)](#), described in Section 3, to assess whether counties that won a Million Dollar Plant (MDP) experienced differential mortality trends relative to the finalist counties that narrowly lost. The design provides the source of identifying variation. I compare winners to their revealed second-choice runners-up and estimate treatment effects using standard difference-in-differences estimators. I report three sets of estimates: a static difference-in-differences specification, estimates based on the heterogeneity-robust estimator of [Callaway and Sant’Anna \(2021\)](#), and a dynamic (event-study) version. Section 7 presents additional robustness checks.

I begin with a static difference-in-differences specification:

$$Y_{ctj} = \beta_1 \mathbb{1}\{\text{Winner}_{cj}\} + \beta_2 \mathbb{1}\{\text{Post}_{tj}\} + \beta_3 (\mathbb{1}\{\text{Post}_{tj}\} \times \mathbb{1}\{\text{Winner}_{cj}\}) + \rho_c + \lambda_t + \eta_j + \mu_{ctj}, \quad (1)$$

where c indexes counties, t years, and j plant-location cases. Y_{ctj} is the crude mortality rate per 100,000 population in county c and year t for a given cause of death. I estimate Equation 1 separately for overdoses, suicides, and alcoholic liver disease, and for demographic subgroups. $\mathbb{1}\{\text{Winner}_{cj}\}$ equals one if county c won the competition associated with case j ; $\mathbb{1}\{\text{Post}_{tj}\}$ equals one in years after the plant opening. The coefficient of interest, β_3 , is the average post-opening difference in mortality between winning and finalist

losing counties. The specification includes county (ρ_c), year (λ_t), and case (η_j) fixed effects, which absorb time-invariant county characteristics, common annual shocks, and average differences across competitions. Following [Greenstone and Moretti \(2003\)](#), the case fixed effects restrict identification to within-competition comparisons between the winner and its finalist losers. Standard errors are clustered at the county level, and observations are weighted by county population.

I also estimate a dynamic, event-study analogue of Equation 1:

$$Y_{ctj} = \sum_{\tau \neq -1} \beta_{\tau} (\mathbb{1}\{\text{Winner}_{cj}\} \times \mathbb{1}\{t - g_j = \tau\}) + \rho_c + \lambda_t + \eta_j + \mu_{ctj}, \quad (2)$$

where g_j is the year the plant opens in case j and $\tau = t - g_j$ denotes event time, with $\tau = -1$ omitted as the reference period. The coefficients $\{\beta_{\tau}\}$ trace the dynamic path of the winner-loser mortality gap around the opening and provide a visual test for differential pre-trends.

The winner-loser design with case fixed effects restricts identification to within-competition comparisons. Nonetheless, because plant openings occur in different years across cases, the static and event-study specifications in Equations 1-2 are two-way fixed-effects (TWFE) estimators, which can be biased when treatment timing is staggered and treatment effects are heterogeneous across cohorts or over time ([Goodman-Bacon, 2021](#); [de Chaisemartin and D’Haultfoeulle, 2020](#)). I therefore also estimate treatment effects using the estimator of [Callaway and Sant’Anna \(2021\)](#), which is robust to such heterogeneity. Counties are grouped into cohorts according to the year in which they first receive treatment (the plant opening). Let G_{cj} denote the cohort of county c in case j . The estimator’s building block is the group-time average treatment effect,

$$ATT(g, t) = E[Y_t(1) - Y_t(0) \mid G = g], \quad (3)$$

the average effect at calendar time t for the cohort first treated in year g . Each $ATT(g, t)$ is estimated as a difference-in-differences contrast between cohort g and a comparison group of not-yet-treated counties, so that no already-treated units enter the control group. The individual $ATT(g, t)$ estimates are then aggregated in two ways. First, an event-study aggregation averages $ATT(g, t)$ across cohorts at each event time τ ,

$$\theta^{CS}(\tau) = \sum_g \omega_g ATT(g, g + \tau), \quad (4)$$

with weights ω_g proportional to cohort size; this yields the dynamic path reported in the event-study figures. Second, an overall aggregation averages the post-treatment $ATT(g, t)$ into a single summary effect,

$$\overline{ATT} = \sum_g \sum_{t \geq g} \omega_{g,t} ATT(g, t), \quad (5)$$

which is the analogue of β_3 in Equation 1 and is the estimate reported in the tables. The Callaway-Sant’Anna estimates serve two purposes: they provide a summary treatment effect that is robust to heterogeneous and dynamic effects under staggered adoption, and, by agreeing with the TWFE estimates, they show that the staggered-timing critique does not overturn the baseline results.

Finally, two remarks on the specifications are in order. First, following [Greenstone and Moretti \(2003\)](#), I deliberately exclude time-varying economic covariates (such as per-capita income, employment, or health-care investment) from all specifications. Because plant openings plausibly affect these variables, they are potential *mechanisms* rather than confounders; controlling for them would absorb part of the treatment effect and bias the estimated total impact of plant openings on mortality. Second, identification throughout rests on the comparability of winners and finalist losers established by the design; Table A.3 shows no statistically significant pre-opening differences between the two groups in mortality, local economic conditions, or educational composition.

6 Results

Table A.4 reports the Average Treatment Effects on the Treated (ATTs) on overdose mortality rates for the overall population (Column 1) and demographic subgroups by gender (Columns 2-3), age (Columns 4-6), and race (Columns 7-9). Panel A shows coefficients from the static Difference-in-Differences (DiD) model in Equation 1, with county, case, and year fixed effects as in [Greenstone and Moretti \(2003\)](#). Panel B applies the estimator of [Callaway and Sant’Anna \(2021\)](#) (Equation 5). Across both specifications, the coefficients are consistently negative. The preferred estimates in Panel B indicate that winning a plant reduces overdose mortality by 2.60 per 100,000 population per year in treated counties, a reduction of 23.3% relative to the sample mean. The reductions are concentrated among men and midlife adults, whose overdose death rates decline by 4.01 (25.5%) and 4.50 (27.3%), respectively. Tables A.5 and A.6 in the Appendix report equivalent estimates for deaths associated with suicide and alcoholic liver disease, showing that the effects are driven almost entirely by reductions in fatalities associated with overdose. Consequently, the remainder of the analysis focuses on these deaths.

Figure A.5 provides dynamic treatment effects using Callaway and Sant’Anna (2021). Panel A.5a examines the effect on the overall population. Coefficients prior to treatment are statistically indistinguishable from zero, indicating an absence of pre-treatment trends. Post-treatment, the coefficients become progressively more negative, suggesting that the impact of the MDPs persisted and intensified over time. Panels A.5b and A.5c explore the effects by gender, revealing that the reduction is more precise and pronounced for men. Panels A.5d-A.5f disaggregate the results by age group, showing that the major reductions occur among midlife adults, followed by older individuals, although the latter estimates are less precise. Finally, Panels A.5g-A.5i break down the effects by race, suggesting that the observed reductions are primarily driven by White individuals. Nevertheless, it is important to note that, since Black and other-race individuals account for around 10% and 1% of the total population, respectively, their estimates are expected to be noisier due to small sample sizes.

Figure A.6 restricts attention to the most affected demographic cells: White men, midlife men, and White midlife men. In line with previous results, Panels A.6a-A.6c show a negative impact of the policy on mortality, which becomes progressively more negative. These results reinforce that the impact of the policy was concentrated among the White population, and in particular White males. Several factors may explain why the relationship between plant openings and mortality is strongest among White males. In particular, a deeper look into the composition of the manufacturing workforce, as well as the profiles most affected by the rise in Deaths of Despair, provides some explanation. First, 76.8% of plants in the sample operated within the manufacturing sector, and according to U.S. Bureau of Labor Statistics (1999), males accounted for 68% of U.S. manufacturing employment in 1999. Second, the population of non-White racial groups is more geographically concentrated than that of Whites, which may reduce the precision of the estimated relationships between the MDPs and mortality rates for those groups. Third, as discussed previously in Section 2, Deaths of Despair are concentrated among middle-aged White Non-Hispanic Americans, and thus the major reversals in deaths should logically be found within these categories.

In summary, the estimations consistently show that the effect of the plant openings grows over time. Coefficients increase in magnitude and significance in the long run, indicating that the impact of the MDPs persists and intensifies over time. This finding carries important policy implications, suggesting that the benefits of some labor-market interventions may take years to fully materialize. Moreover, the delayed manifestation of these effects aligns with the nature of Deaths of Despair, which often represent the culmination of prolonged periods of suffering and mental distress. Large plant openings are unlikely to produce immediate reductions in mortality; instead, they may help mitigate the underlying factors contributing to such deaths, including feelings of loneliness, failure, or social disconnection among vulnerable population subgroups. As

a result, we can anticipate that the observed effects begin to materialize in the medium term and gradually intensify over time.

6.1 Mechanisms

To explore whether the labor-market improvements generated by Million Dollar Plants can account for the reduction in overdose mortality, I first establish that winning a plant raised local economic activity, and then use a placebo exercise to assess whether the mortality effect reflects a broad improvement in health rather than the specific deaths-of-despair channel.

Table A.7 and Figure A.7 report the effect of plant openings on local labor markets. Winning a plant increased county employment by roughly 10 percent, the number of establishments by about 8 percent, and total wages by 9-10 percent. The static difference-in-differences and Callaway-Sant’Anna estimates are almost identical (for example, 0.097 versus 0.098 for log employment), and the event-study estimates display no differential pre-trends, with effects emerging after the opening and persisting thereafter. These results confirm that the winner-runner-up comparison captures a meaningful, positive shock to local labor-market conditions, the first stage through which plant openings could plausibly affect mortality.

Labor markets can influence mortality through several channels, including improved access to health insurance, reduced economic stress, and healthier behaviors. The relationship between economic conditions and health has been widely documented (Cutler et al., 2006). In this setting, the observed decline in deaths of despair in winning counties could reflect either a reduction in the behaviors that drive these specific causes (such as smoking, alcohol consumption, or substance use) or a more general improvement in health and access to healthcare. Distinguishing between these interpretations is important: the former would tie the effect to the deaths-of-despair channel, while the latter would suggest that winning a plant simply made counties healthier across the board.

To separate these interpretations, I test whether plant openings affected mortality from twelve categories of causes unrelated to deaths of despair, such as blood, endocrine, respiratory, and digestive disorders, which are not plausibly linked to the economic-stress or behavioral channels but would respond to a general improvement in health or healthcare access. Figure A.8 reports these placebo estimates. Across the twelve categories, I find no systematic relationship between plant openings and mortality. This is informative: it indicates that winning a plant did not produce a broad reduction in mortality from other causes, which rules out a general improvement in health conditions or healthcare access as the explanation for the main results. The effect of plant openings is therefore concentrated in deaths of despair, consistent with a channel operating through economic conditions and the behaviors associated with these specific causes, rather than

through population health more broadly.

7 Robustness Checks

In this section, I conduct a series of robustness checks to evaluate the sensitivity of the main findings to alternative specifications. Specifically, I address concerns that the observed effects could be attributed to the differential regional timing of the opioid epidemic, diverging trends in opioid supply, or differences in state-level opioid regulations. First, I add region- and state-by-year fixed effects to absorb differential regional trends in the epidemic. Second, I test whether winner and loser counties exhibit diverging trends in opioid supply. Third, I account for variation in state-level opioid regulations. These analyses confirm that the main findings are robust and not driven by these alternative explanations.

7.1 Regional Timing of the Opioid Epidemic

A potential concern is that the estimated effects reflect the differential timing of the opioid epidemic across regions rather than the plant openings themselves. Because the epidemic diffused across regions at different times, winner and loser counties could have diverged for reasons unrelated to the plants. To address this, I modify the static difference-in-differences specification of Equation 1 by adding progressively more demanding fixed effects: region-by-year and, in the most stringent specification, state-by-year fixed effects, which absorb any shock common to a region (or state) in a given year, including the local trajectory of the epidemic. This modification retains the county, case, and year fixed effects of Equation 1, so identification continues to rest on within-competition comparisons between a winner and its finalist losers. Because fixed-effects absorption of this kind is not available within the Callaway-Sant’Anna framework, these estimates modify the two-way fixed-effects estimator of Equation 1 rather than the Callaway and Sant’Anna specification.

Table A.9 reports the results. The estimated effect is robust to the inclusion of region-by-year fixed effects, with point estimates close to the baseline. Under the most demanding state-by-year specification, the point estimate remains similar in magnitude but is less precisely estimated, reflecting the substantial identifying variation absorbed by state-year fixed effects. Figures A.11 and A.12 show the corresponding event studies with region-by-year fixed effects, which confirm the absence of differential pre-trends. Overall, the results indicate that the estimated reduction in overdose mortality is not an artifact of differential regional epidemic timing.

7.2 Testing for Differential Trends in the Supply of Opioids

Several works in the literature show that the rise in drug-overdose deaths can be tracked through the increase in the supply and availability of opioids (Alpert et al., 2022; Cutler and Glaeser, 2021; Ruhm, 2018). In particular, if loser counties systematically experienced greater increases in opioid supply, the results presented in the previous section could be spuriously driven by differences in drug availability rather than by the plant openings. This concern is particularly salient given the substantial variation in state laws regulating opioid supply and prescribing practices (Meara et al., 2016; Morden et al., 2014). To address this issue, I use data on opioid shipments from ARCOS reports, covering 1997-2017. Details of these data are provided in Section 4. Because 1997 is the earliest year available, I cannot use opioid supply as a control variable in the baseline regressions; instead, I test whether supply differs between winner and loser counties.

Figure A.13 shows boxplots comparing the distribution of total opioid supply, by year, between winner and loser counties, with two-sample t -tests of differences in means marked above each pair. The results show no systematic differences in opioid supply between winner and loser counties in any year. Figure A.14 repeats the exercise separately for each drug category (oxycodone, hydrocodone, fentanyl, morphine, and codeine), and I again find no evidence of differential supply for any component in any year. Table A.10 confirms this in a pooled regression: the winner-versus-loser difference in supply is small and statistically insignificant for total supply and for every drug, indicating that opioid availability did not differ systematically between the two groups of counties.

7.3 Controlling for Differences in State Legislation of Opioids

To assess whether differences in opioid regulations could explain the observed reductions in deaths of despair, I use data on state-level policy variation compiled by Meara et al. (2016), which documents legislative actions related to opioid regulation, including prescribing limits, patient-monitoring requirements, and pain-clinic regulations, across all U.S. states from 2006 to 2012. More details are provided in Section 4. Following Pierce and Schott (2020), I construct a measure of regulatory stringency equal to the number of opioid-related legislative categories in effect in each state, and I interact this measure with year dummies in the static difference-in-differences specification (Equation 1).

Table A.8 and Figures A.9 and A.10 display the results. The estimates are robust to the inclusion of the opioid-legislation controls: the magnitude and significance of the coefficients are largely unchanged relative to the baseline, indicating that the reduction in overdose mortality in winning counties is not explained by differences in state opioid regulation.

8 Conclusion

In 1893, Émile Durkheim explored the implications of labor on social cohesion in his seminal work, *The Division of Labor in Society* (Durkheim, 1893). He posited that occupational routines play a vital role in fostering social cohesion, as they connect individuals to collective purposes and shared norms. When these connections are disrupted, society can fall into a state of “anomie,” or normlessness, where individuals lose their sense of belonging and shared direction. Durkheim elaborated on this concept in his later work, *Suicide: A Study in Sociology* (Durkheim, 1897). In this work, he demonstrated how economic downturns and disruptions in work life could precipitate “anomic” suicide, as individuals lose the regulatory frameworks and collective purposes provided by stable employment. Durkheim’s motivation for these studies was driven by the rapid societal changes of the late nineteenth century, including industrialization and urbanization. These upheavals disrupted traditional forms of social cohesion, as individuals became detached from shared norms and collective purposes that once provided stability. Durkheim’s work stands as a cautionary tale, urging us to assess both the social and economic implications of economic development.

This paper has examined the relationship between county-level mortality rates and labor market conditions, leveraging the opening of Million Dollar Plants (MDPs) between 1982 and 1993. These plant openings generated improved employment opportunities in selected US counties, thereby providing a source of exogenous variation to examine how better labor market prospects influence Deaths of Despair. Using a staggered Difference-in-Differences framework, I find that (1) the opening of these plants significantly reduced overdose mortality; (2) the effects are most pronounced among middle-aged adults, particularly middle-aged men; (3) the impact intensifies over time, indicating a persistent and cumulative benefit; and (4) the effect is specific to overdose deaths, with no significant change in suicide or alcoholic liver disease mortality. To the best of my knowledge, this is the first study to establish a causal link between improved labor market conditions and reduced Deaths of Despair. Prior research has predominantly focused on the negative effects of job loss rather than the positive effects of employment on these outcomes. In doing so, the paper also speaks directly to supply-side skepticism about place-based policy: Ruhm (2018) argued that “*efforts to improve economic conditions in distressed locations, while desirable for other reasons, are unlikely to yield significant reductions in drug mortality.*” The evidence presented here points in the opposite direction. This paper therefore makes two key contributions. First, it provides insights that could inform policy interventions aimed at mitigating the rising mortality associated with substance abuse in the United States. Second, while not explicitly testing Durkheim’s theory, the findings resonate with his views on the critical role of labor markets in shaping societal distress. These results emphasize the broader implications of labor market dynamics for social cohesion and public health, areas often overlooked in economic analyses but deeply rooted in other

disciplines.

These findings also speak to how large an economic intervention is needed to move mortality of this kind, and to the policy calculus of doing so. It is notable that the mortality reduction from a positive employment shock is smaller than the increase [Venkataramani et al. \(2020\)](#) document from plant closures (2.60 versus 8.6 deaths per 100,000). This asymmetry echoes both the behavioral notion of loss aversion and Durkheim’s emphasis on disruption: the tearing apart of stable work appears to inflict more harm than its creation repairs. The policy calculus is nonetheless favorable. The same plant openings have been shown to raise local productivity and property values ([Greenstone and Moretti, 2003](#)), so the reduction in overdose mortality documented here is best understood as an additional, previously unmeasured benefit of a place-based policy that communities were already willing to fund on conventional economic grounds.

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A Appendix

A.1 Figures

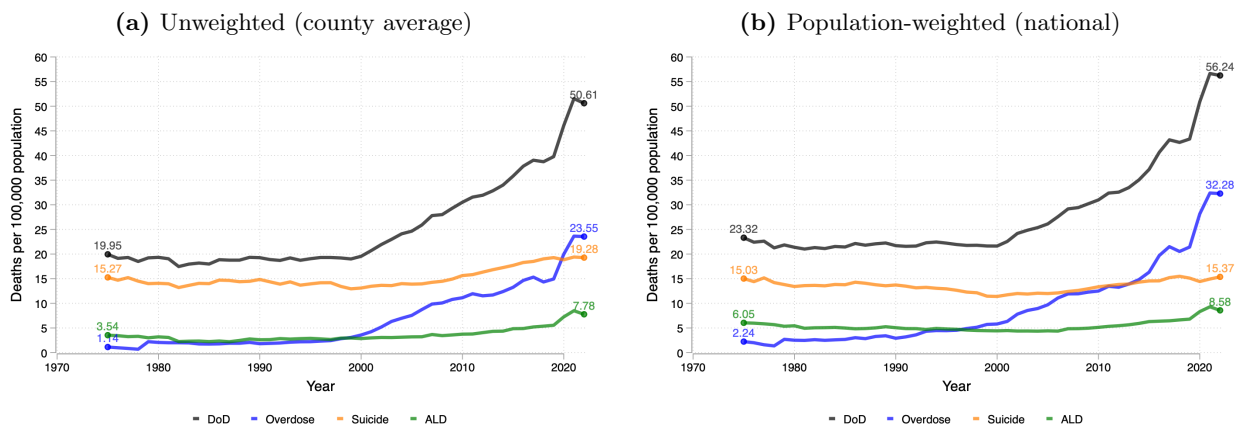


Figure A.1: Evolution of Deaths-of-Despair Mortality, 1975-2022

Notes: Deaths of despair refer to deaths caused by drug overdose, suicide, and alcoholic liver disease. The figure displays crude mortality rates at the national level, 1975-2022, calculated as deaths per 100,000 population, for total deaths of despair and each of its three components. The left panel reports the unweighted average across U.S. counties and the right panel the population-weighted national rate; endpoint values are labeled. Death counts are compiled by the author from National Center for Health Statistics (NCHS) microdata; population data are from the Surveillance, Epidemiology, and End Results (SEER) Program.

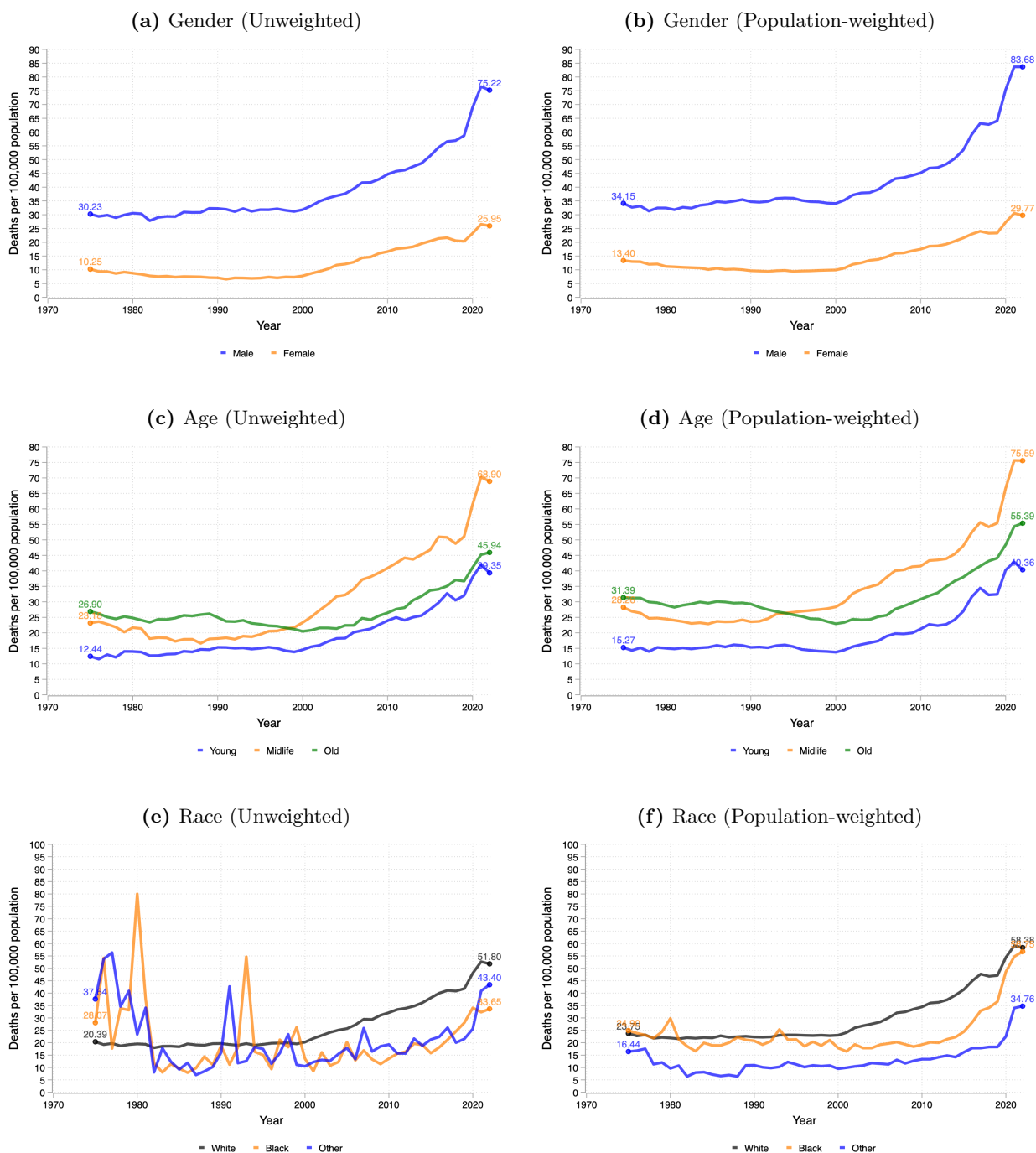


Figure A.2: Evolution of Deaths-of-Despair Mortality by Demographic Group, 1975-2022

Notes: Deaths of despair refer to deaths caused by drug overdose, suicide, and alcoholic liver disease. The figure displays crude deaths-of-despair mortality rates at the national level, 1975-2022, calculated as deaths per 100,000 population, disaggregated by gender (top), age group (middle; young 15-34, midlife 35-54, old 55+), and race (bottom). Within each row, the left panel reports the unweighted average across U.S. counties and the right panel the population-weighted national rate; endpoint values are labeled. Death counts are compiled by the author from National Center for Health Statistics (NCHS) microdata; population data are from the Surveillance, Epidemiology, and End Results (SEER) Program.

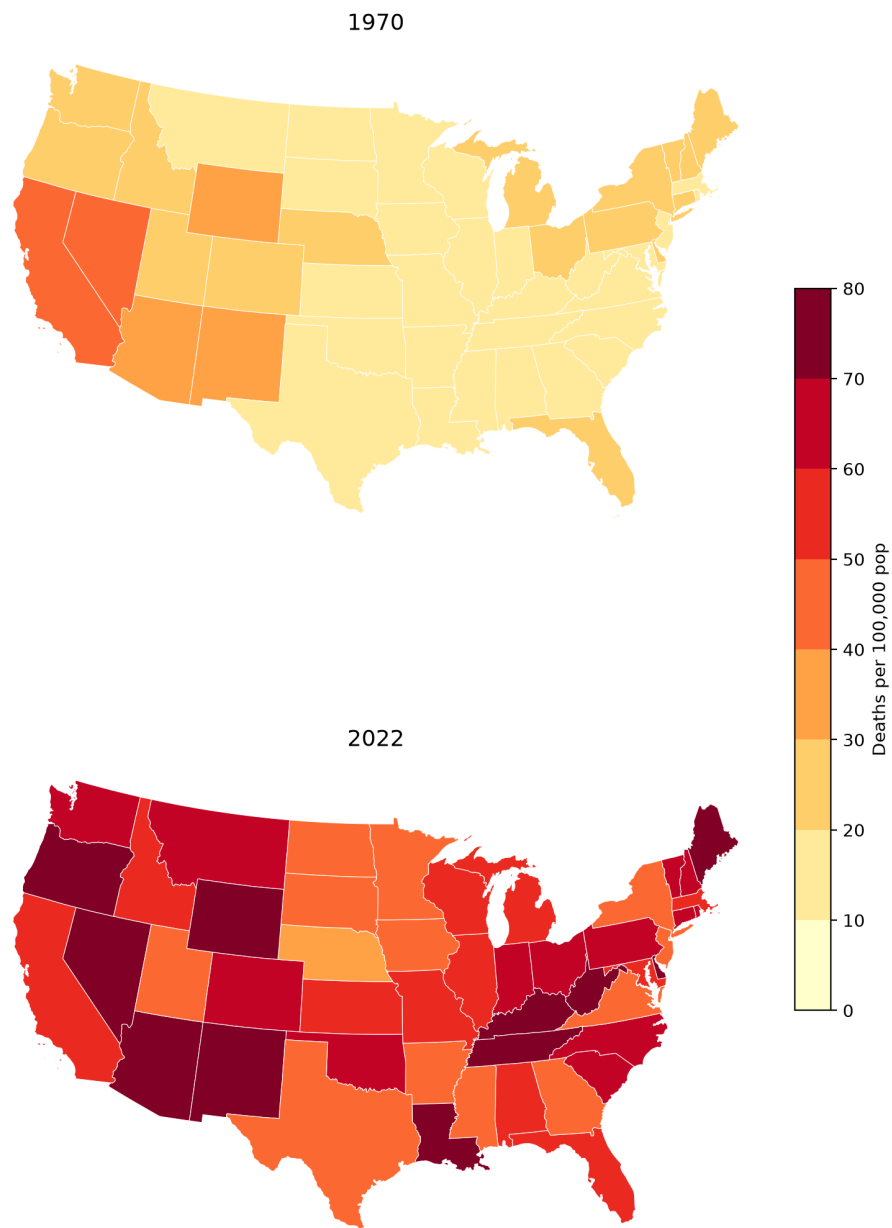


Figure A.3: Deaths of Despair by State, 1970 vs. 2022

Notes: Population-weighted crude deaths-of-despair mortality rate (deaths per 100,000) by state, in 1970 (top) and 2022 (bottom). State rates are computed as the population-weighted average of county rates. Both panels use a common scale to make the two years comparable. Death counts are compiled by the author from National Center for Health Statistics (NCHS) microdata; population data are from the Surveillance, Epidemiology, and End Results (SEER) Program. Alaska and Hawaii are omitted.

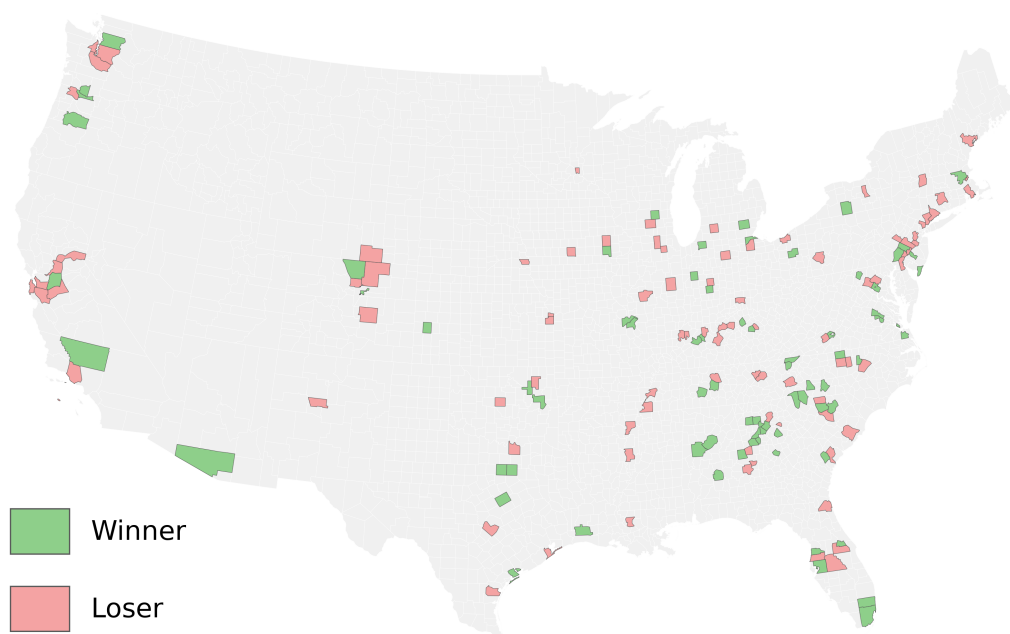


Figure A.4: Winner and Loser Counties in the Million Dollar Plants Sample

Notes: Counties in the estimation sample. Winner counties (those that received a large manufacturing plant) are shaded green; loser (finalist runner-up) counties are shaded red. Sample constructed from the Million Dollar Plants case reports following Greenstone and Moretti (2003), after the exclusions described in Section 3. Alaska and Hawaii are omitted.

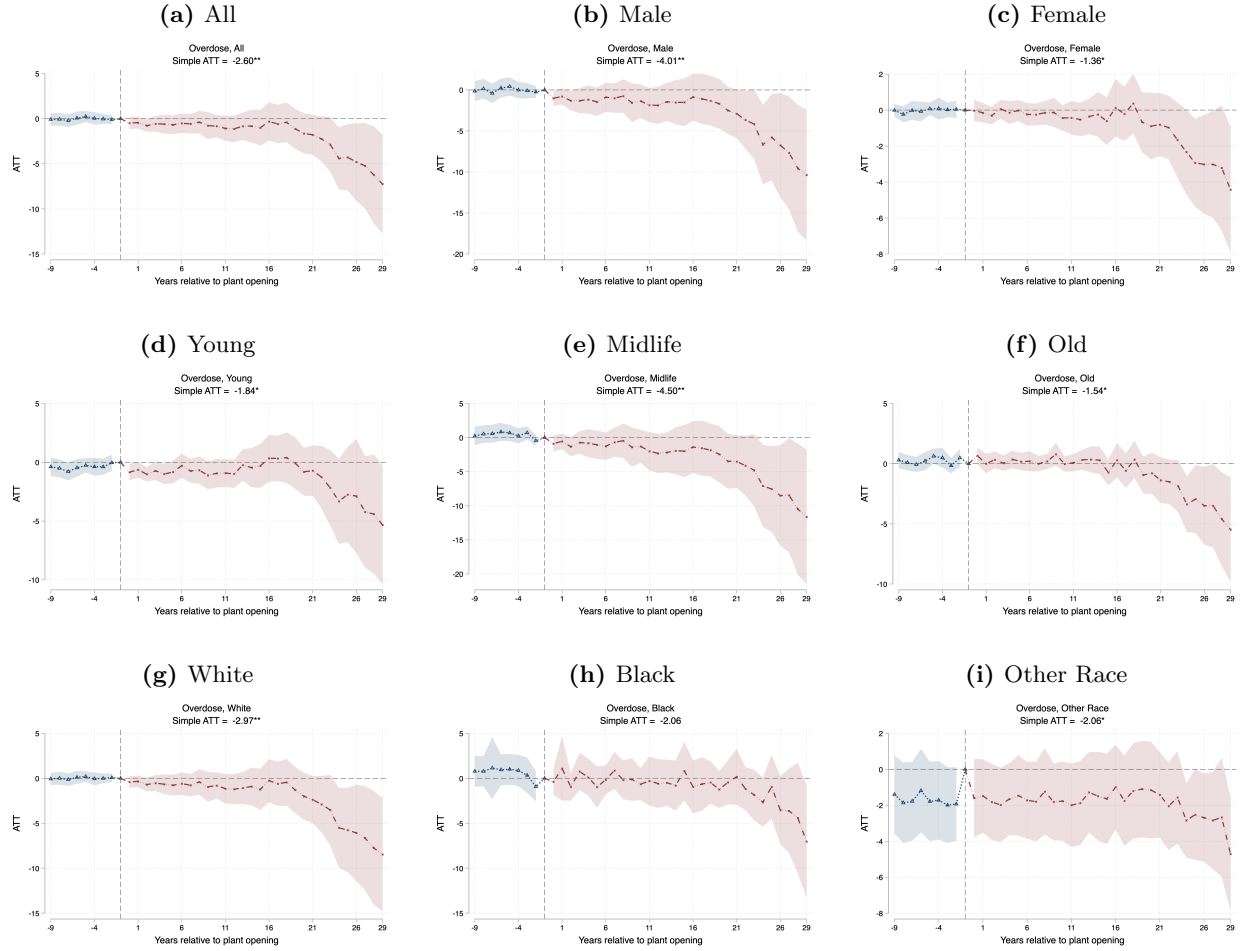


Figure A.5: Dynamic Effects of Million Dollar Plant Openings on Overdose Mortality: Callaway-Sant'Anna Event-Study Estimates

Notes: This figure reports Callaway and Sant'Anna (2021) event-study estimates of the effect of winning a large manufacturing plant on overdose mortality (deaths per 100,000), for the full population and by gender, age, and race. Event time is measured relative to plant opening, with $t = -1$ omitted as the reference period; the balanced window spans -9 to $+29$. Lines denote coefficient estimates and shaded areas represent 95 percent confidence intervals based on county-clustered standard errors. Negative coefficients indicate lower overdose mortality in winner counties relative to finalist losing counties. Each panel's subtitle reports the pooled ATT.

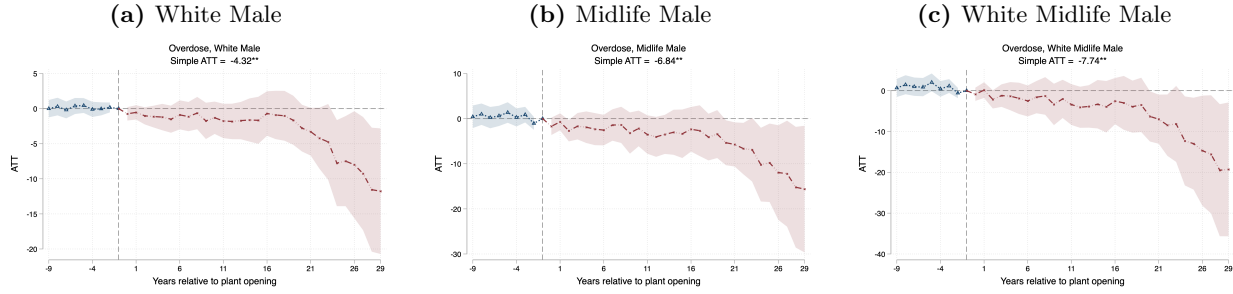


Figure A.6: Dynamic Effects of Million Dollar Plant Openings on Overdose Mortality: Additional Demographic Subgroups, Callaway-Sant'Anna

Notes: This figure reports Callaway and Sant'Anna (2021) event-study estimates for White males, midlife males, and White midlife males. Event time is measured relative to plant opening, with $t = -1$ omitted as the reference period. Lines denote coefficient estimates and shaded areas represent 95 percent confidence intervals based on county-clustered standard errors. Negative coefficients indicate lower overdose mortality in winner counties relative to finalist losing counties.

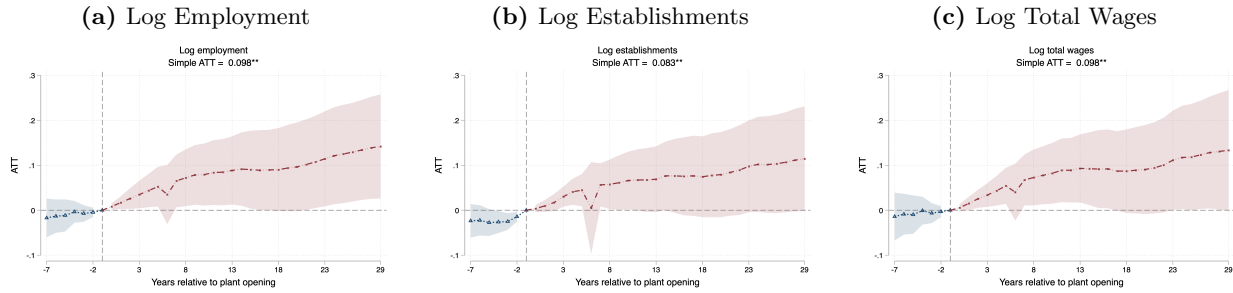


Figure A.7: Dynamic Effects of Million Dollar Plant Openings on Local Labor Markets: Callaway-Sant'Anna Event-Study Estimates

Notes: This figure reports Callaway and Sant'Anna (2021) event-study estimates of the effect of winning a large manufacturing plant on log employment, log establishments, and log total wages (QCEW). Event time is measured relative to plant opening, with $t = -1$ omitted as the reference period; the balanced window spans -7 to $+29$ (QCEW data begin in 1975). Lines denote coefficient estimates and shaded areas represent 95 percent confidence intervals based on county-clustered standard errors. Positive coefficients indicate higher values in winner counties relative to finalist losing counties. Each panel's subtitle reports the pooled ATT.

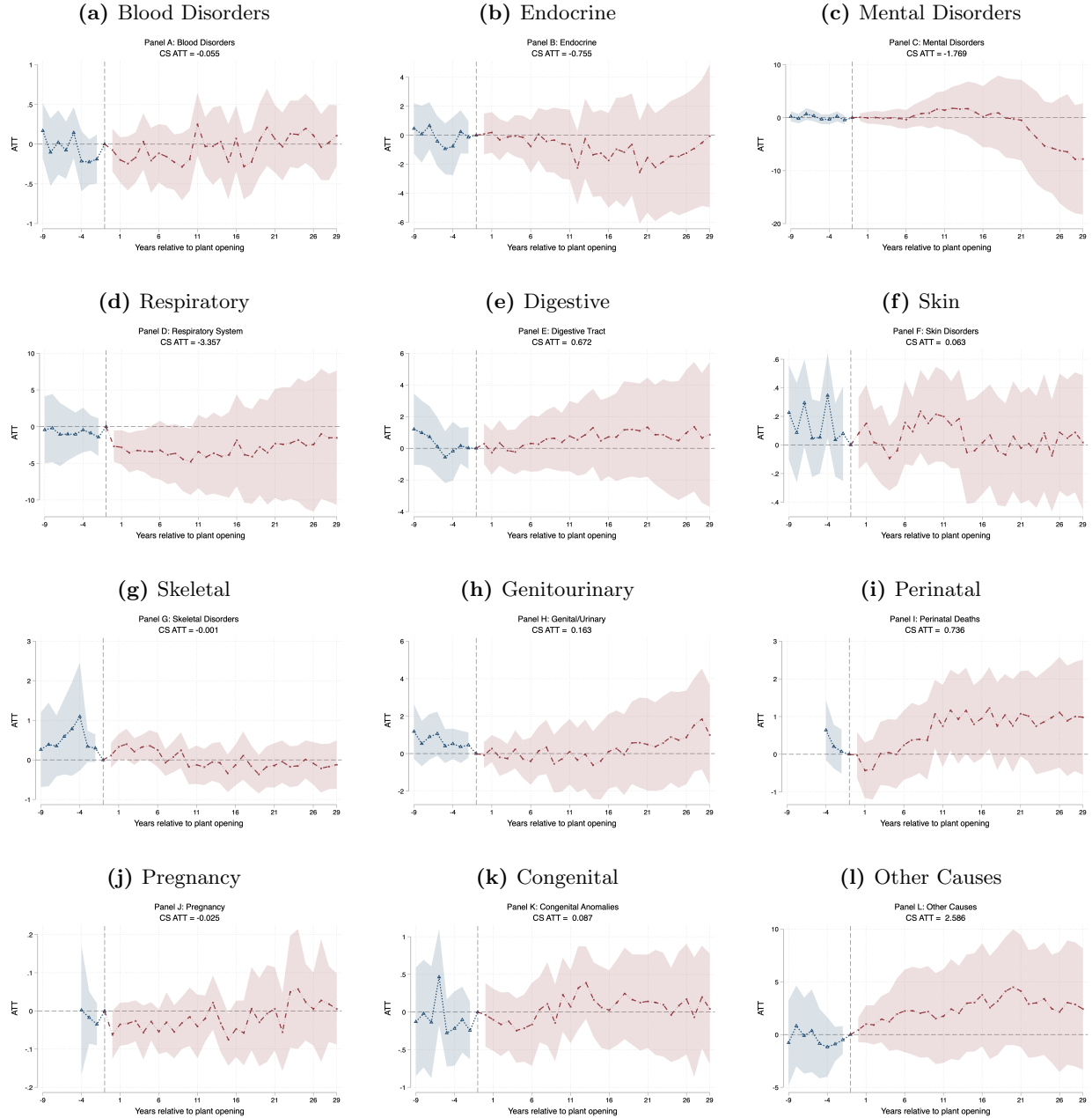


Figure A.8: Placebo Event Studies: Effect of Plant Openings on Non-Deaths-of-Despair Mortality
Notes: Callaway and Sant’Anna (2021) event-study estimates of the effect of winning a large manufacturing plant on mortality from causes unrelated to deaths of despair (placebo outcomes), per 100,000 population. Event time is relative to plant opening, with $t = -1$ omitted as the reference period. Lines are point estimates; shaded areas are 95% confidence intervals based on county-clustered standard errors. The absence of systematic effects supports the interpretation that the estimated reduction in overdose mortality reflects the plant’s effect rather than broad differences between winner and loser counties.

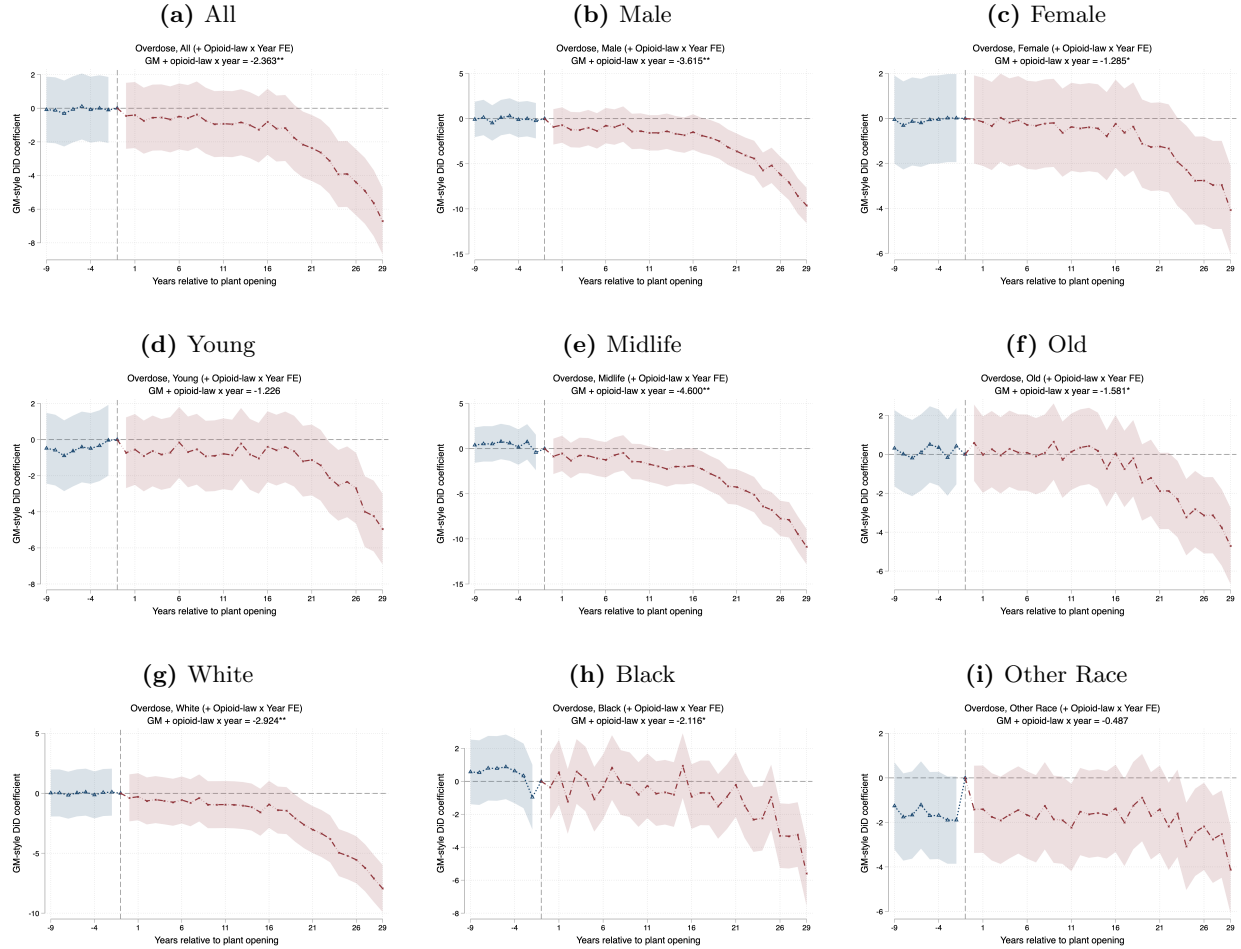


Figure A.9: Dynamic Effects on Overdose Mortality, Controlling for State Opioid Legislation: Two-Way Fixed-Effects Event-Study Estimates

Notes: This figure reports two-way fixed-effects (Greenstone-Moretti style) event-study estimates of the effect of winning a large manufacturing plant on overdose mortality (deaths per 100,000), for the full population and by gender, age, and race, interacting the state opioid-legislation measure (Meara et al. 2016) with year fixed effects. Event time is measured relative to plant opening, with $t = -1$ omitted as the reference period. Lines denote coefficient estimates and shaded areas represent 95 percent confidence intervals based on county-clustered standard errors. Negative coefficients indicate lower overdose mortality in winner counties relative to finalist losing counties. Each panel's subtitle reports the pooled effect.

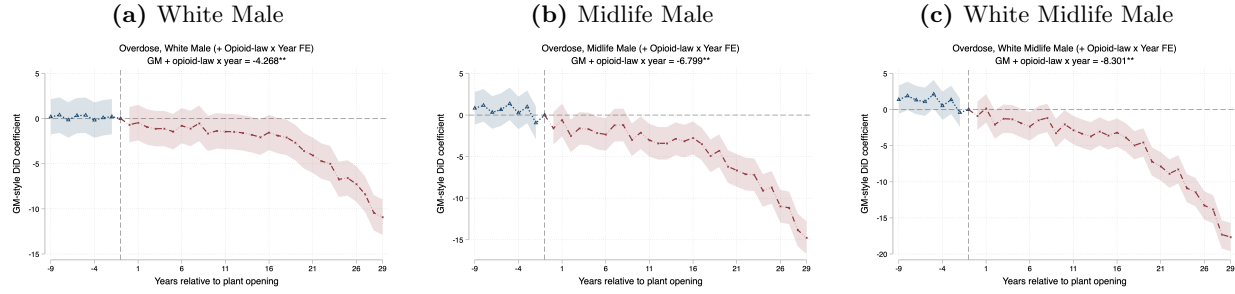


Figure A.10: Dynamic Effects on Overdose Mortality, Controlling for State Opioid Legislation: Additional Demographic Subgroups

Notes: Two-way fixed-effects (Greenstone-Moretti style) event-study estimates for White males, midlife males, and White midlife males, interacting the state opioid-legislation measure (Meara et al. 2016) with year fixed effects. Event time is measured relative to plant opening, with $t = -1$ omitted as the reference period. Lines denote coefficient estimates and shaded areas represent 95 percent confidence intervals based on county-clustered standard errors. Negative coefficients indicate lower overdose mortality in winner counties relative to finalist losing counties.

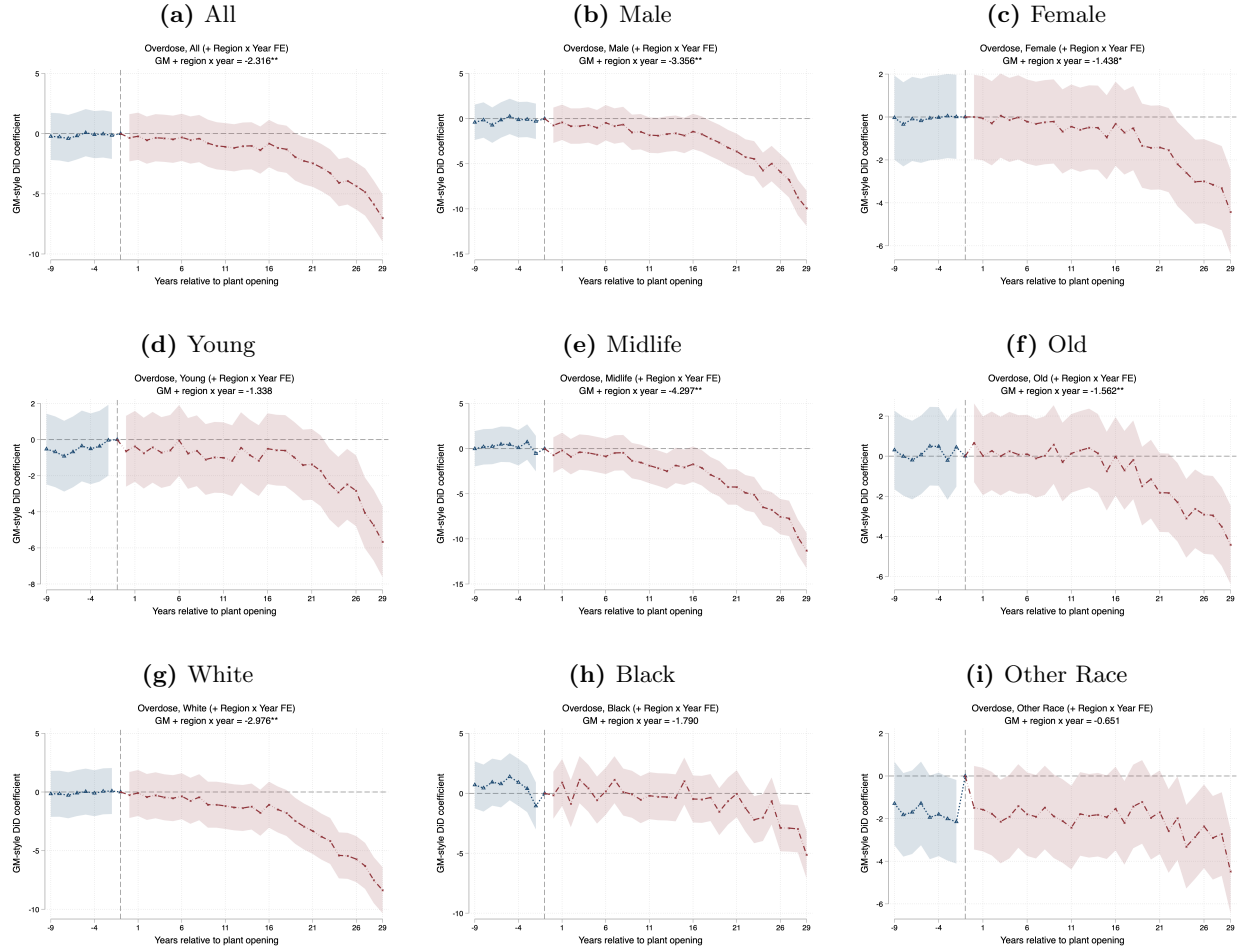


Figure A.11: Dynamic Effects on Overdose Mortality with Region-by-Year Fixed Effects

Notes: This figure reports two-way fixed-effects event-study estimates of the effect of winning a large manufacturing plant on overdose mortality (deaths per 100,000), for the full population and by gender, age, and race, adding region-by-year fixed effects to absorb differential regional timing of the opioid epidemic. Event time is measured relative to plant opening, with $t = -1$ omitted as the reference period. Lines denote coefficient estimates and shaded areas represent 95 percent confidence intervals based on county-clustered standard errors. Region-by-year fixed effects are not available within the Callaway-Sant’Anna framework, so these estimates use the two-way fixed-effects estimator.

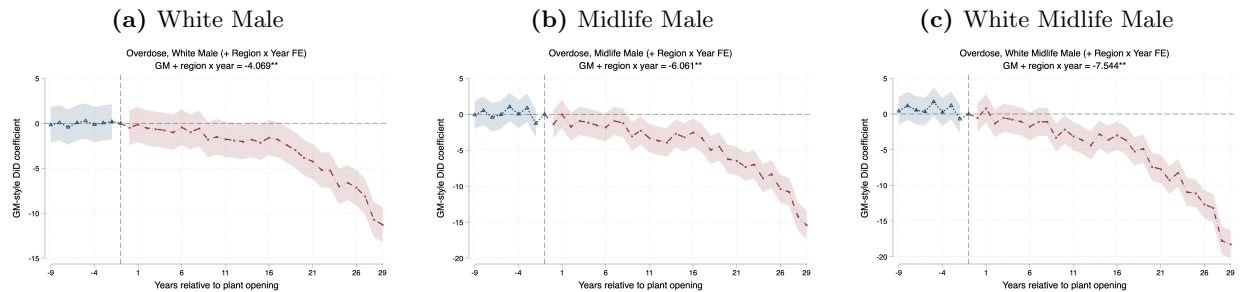


Figure A.12: Dynamic Effects on Overdose Mortality with Region-by-Year Fixed Effects: Additional Demographic Subgroups

Notes: Two-way fixed-effects event-study estimates for White males, midlife males, and White midlife males, adding region-by-year fixed effects. Event time is measured relative to plant opening, with $t = -1$ omitted as the reference period. Shaded areas are 95 percent confidence intervals based on county-clustered standard errors.

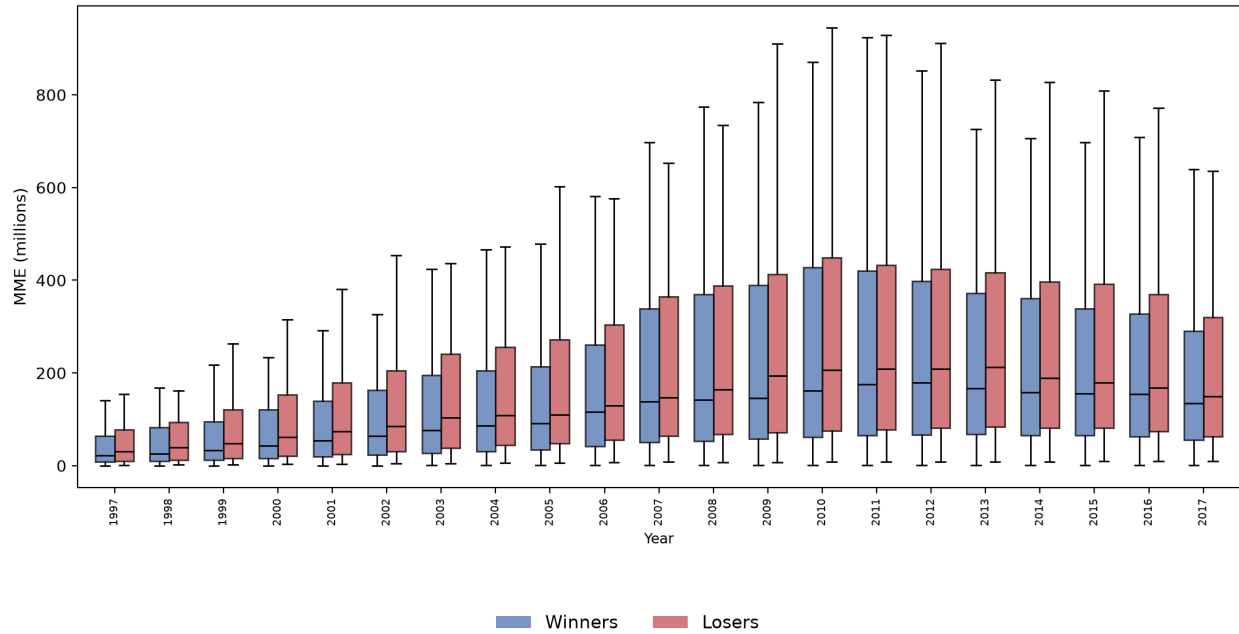


Figure A.13: Distribution of Total Opioid Supply in Winner and Loser Counties, by Year (1997-2017)

Notes: Each boxplot shows the distribution across counties of total opioid supply (in morphine milligram equivalents, MME, summed over oxycodone, hydrocodone, fentanyl, morphine, and codeine) for a given year, 1997-2017. Winner counties are in blue and loser counties in red; outliers are omitted for readability. Two-sample t -tests compare mean supply between winner and loser counties each year, with significance levels marked above each pair. The absence of significant differences supports the identifying assumption that estimated mortality effects are not driven by differential opioid availability. Supply data are from the DEA Automation of Reports and Consolidated Orders System (ARCOS).

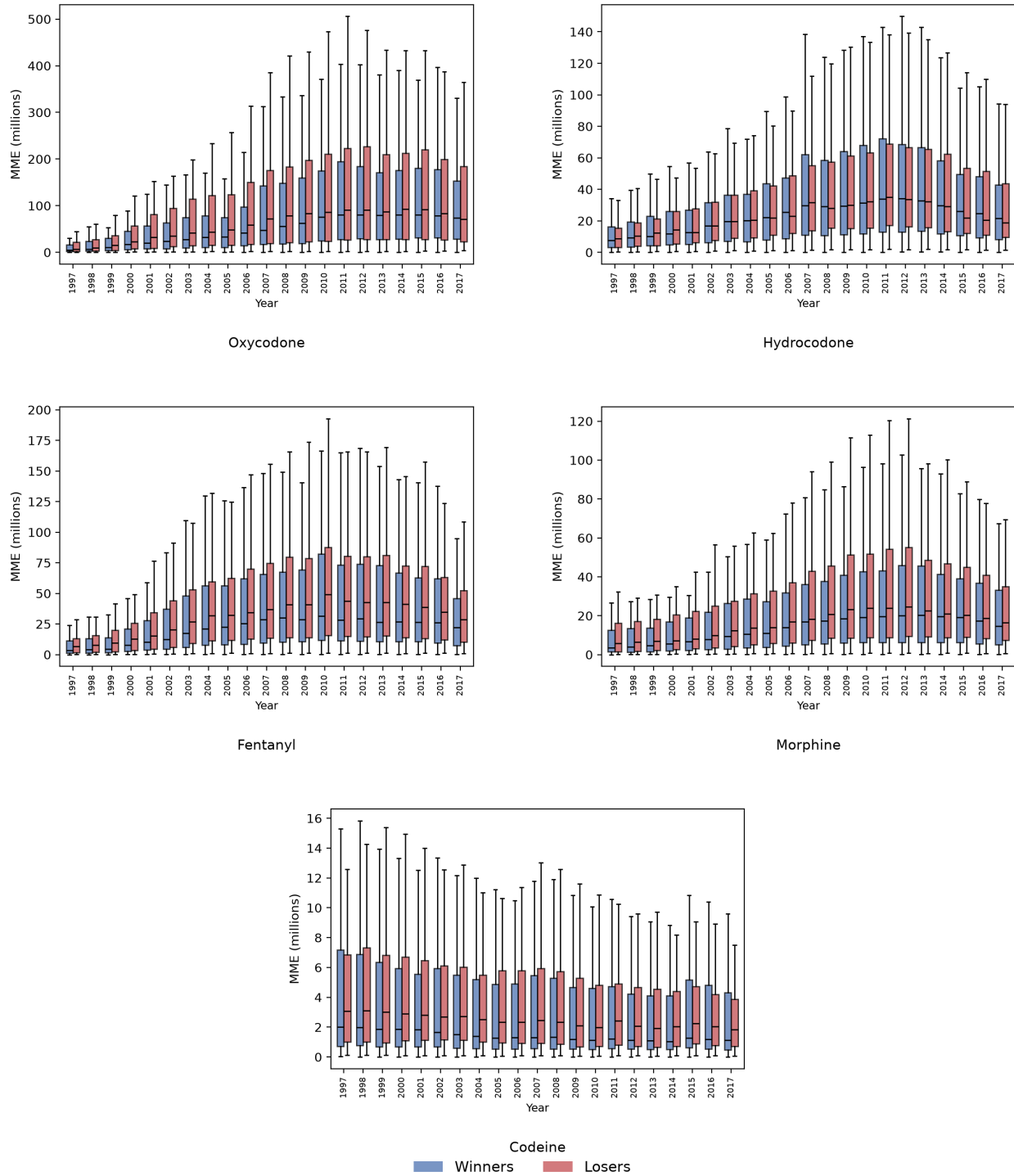


Figure A.14: Distribution of Opioid Supply in Winner and Loser Counties, by Type and Year (1997-2017)
Notes: Each boxplot shows the distribution across counties of opioid supply (MME) for a given drug and year: oxycodone, hydrocodone, fentanyl, morphine, and codeine. Winner counties are in blue and loser counties in red. Two-sample *t*-tests compare mean supply between winner and loser counties each year; significance levels are marked above each pair. No significant differences are found for any drug in any year.

A.2 Tables

Table A.1: Characteristics of the Million Dollar Plants Sample

	Original	Analysis
<i>Sector (% of cases)</i>		
Manufacturing	76.8	77.5
Services	7.3	7.5
Trade	4.9	5.0
Transportation & Utilities	9.8	8.8
Financials	1.2	1.2
<i>Opening year (% of cases)</i>		
1982	6.1	6.2
1983	3.7	3.8
1984	3.7	3.8
1985	7.3	7.5
1986	7.3	7.5
1987	9.8	10.0
1988	11.0	10.0
1989	8.5	8.8
1990	7.3	7.5
1991	14.6	13.8
1992	12.2	12.5
1993	8.5	8.8
<i>Sample size (N)</i>		
Winner counties	82	80
Loser counties	129	108
Total counties	211	188
Cases (plants)	82	80

Notes: This table describes the sample of Million Dollar Plants used in the analysis. The unit of observation is a case (one winning plant per plant-location competition). Sector, opening period, and Census region shares are computed over cases; percentages are of the total number of cases. The sample is constructed from the Million Dollar Plants case reports following Greenstone and Moretti (2003), after the exclusions described in Section 3 (dropping contaminated control counties and cases in which the same county appears as both winner and loser). The final block reports the number of winner and loser counties in the estimation sample.

Table A.2: Switcher Counties: Competitions Won vs. Lost

County	Won (firm, year)	Lost (firm, year)
Denver, CO	United Airlines (1991)	United Airlines (1991)
Larimer, CO	Wal-Mart Stores (1989)	Hewlett-Packard (1983)
Hillsborough, FL	JLM Industries (1993)	GTE (1989)
Dade, FL	MCI Communications (1992)	Racal-Milgo (1982)
DeKalb, GA	Bass (1990); Holiday (1991)	Anheuser-Busch (1988)
Daviess, KY	Scott Paper (1992)	Worlmark (1987)
Middlesex, MA	Codex (Motorola) (1984); Reebok International (1988)	AMF/Reece (1991)
St. Louis County, MO	Everest & Jennings (1992)	General Motors (1984)
Durham, NC	Reichhold Chemicals (1990)	DuPont/Phillips (1986); Mercedes-Benz (1993)
Lucas, OH	Burlington Air Express (1989)	Philips Display (1989)
Montgomery, PA	Ford (1991); Sterling Drug (1993)	Eastman Kodak (1988)
Fairfax, VA	TRW (1985)	American Auto (1991); United Airlines (1991)
Henrico, VA	Wella (1988)	Canon (1986)
Berkeley, WV	Swearingen Aircraft (1992)	United Airlines (1991); AT&T (1992)

Notes: The 14 “switcher” counties each won at least one Million Dollar Plants competition and lost at least one other. Because a county treated in one competition cannot serve as a clean control in another, their losing (control) observations are dropped as contaminated controls (Section 3); their winning observations are retained. Compiled from the Million Dollar Plants case reports.

Table A.3: Baseline Balance: Winner vs. Loser Counties

	Winners	Losers	Difference	p-value
Deaths of despair per 100,000	22.03	22.23	-0.20	0.885
Overdose per 100,000	3.13	3.34	-0.21	0.564
Suicide per 100,000	14.31	14.01	0.30	0.723
Alcoholic liver disease per 100,000	4.59	4.88	-0.29	0.552
Employment-to-population ratio	0.63	0.62	0.01	0.804
Average annual pay (USD)	21644.79	21909.44	-264.66	0.784
Establishments per 1,000 residents	30.72	31.06	-0.34	0.804
% with college degree	22.06	21.08	0.98	0.459

Notes: This table compares winner and loser counties in the three years before the plant-opening announcement ($\tau = -3$ to -1), averaged within each county. Columns report population-weighted means for winner and loser counties, their difference, and the p -value from a population-weighted test of equality (robust standard errors). Overdose and deaths-of-despair rates are per 100,000 population; economic conditions are population-adjusted (employment-to-population ratio, average annual pay, and establishments per 1,000 residents) to capture economic intensity rather than county size; education is the share of adults with a college degree. The unit of observation is a case-county in the pre-opening period. *, **, and *** denote significance of the winner-loser difference at the 10%, 5%, and 1% levels.

Table A.4: Effect of Plant Openings on Overdose Mortality

	All Population	By Gender		By Age Group			By Race Group		
	All	Male	Female	Young	Midlife	Old	White	Black	Other
<i>Panel A: Static DiD with case fixed effects (Greenstone and Moretti, 2003), Overdose</i>									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treated $\times$ After Treatment	-2.49** (1.20)	-3.80** (1.75)	-1.36* (0.75)	-1.35 (1.03)	-4.82** (2.23)	-1.72* (0.88)	-2.98** (1.45)	-2.64** (1.26)	-0.59 (0.70)
Mean	11.15	15.75	6.86	9.51	16.51	7.27	11.97	12.15	3.15
N	9258	9258	9258	9258	9258	9258	9258	9243	9258
<i>Panel B: Callaway-Sant'Anna ATT, Overdose</i>									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treated $\times$ After Treatment	-2.60** (1.11)	-4.01** (1.59)	-1.36* (0.74)	-1.84* (0.96)	-4.50** (2.01)	-1.54* (0.86)	-2.97** (1.34)	-2.06 (1.28)	-2.06* (1.18)
Mean	11.15	15.75	6.86	9.51	16.51	7.27	11.97	12.15	3.15
N	9196	9196	9196	9196	9196	9196	9196	9181	9196

Notes: This table reports the estimated effect of winning a large manufacturing plant on overdose mortality (deaths per 100,000), for the full population and by gender, age, and race. Panel A reports static difference-in-differences estimates with county, case, and year fixed effects, following the Greenstone and Moretti (2003) winner-runner-up design; case fixed effects restrict comparisons to counties competing for the same plant. Panel B reports Callaway and Sant'Anna (2021) estimates robust to staggered adoption. All specifications are population-weighted, with standard errors clustered at the county level. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table A.5: Effect of Plant Openings on Suicide Mortality

	All Population	By Gender		By Age Group			By Race Group		
	All	Male	Female	Young	Midlife	Old	White	Black	Other
<i>Panel A: Static DiD with case fixed effects (Greenstone and Moretti, 2003), Suicide</i>									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treated $\times$ After Treatment	-0.48 (0.61)	-0.85 (0.88)	-0.16 (0.41)	-0.26 (0.60)	-0.77 (0.73)	-0.79 (0.68)	-0.62 (0.71)	-0.93 (0.66)	-0.43 (0.75)
Mean	13.28	21.04	6.09	11.01	14.08	15.15	14.87	7.76	6.26
N	9258	9258	9258	9258	9258	9258	9258	9243	9258
<i>Panel B: Callaway-Sant'Anna ATT, Suicide</i>									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treated $\times$ After Treatment	-0.51 (0.50)	-0.82 (0.83)	-0.19 (0.35)	-0.27 (0.79)	-0.79 (0.67)	-1.11 (0.88)	-0.58 (0.51)	0.40 (0.87)	-0.15 (1.12)
Mean	13.28	21.04	6.09	11.01	14.08	15.15	14.87	7.76	6.26
N	9196	9196	9196	9196	9196	9196	9196	9181	9196

Notes: This table reports the estimated effect of winning a large manufacturing plant on suicide mortality (deaths per 100,000), for the full population and by gender, age, and race. Panel A reports static difference-in-differences estimates with county, case, and year fixed effects, following the Greenstone and Moretti (2003) winner-runner-up design; case fixed effects restrict comparisons to counties competing for the same plant. Panel B reports Callaway and Sant'Anna (2021) estimates robust to staggered adoption. All specifications are population-weighted, with standard errors clustered at the county level. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table A.6: Effect of Plant Openings on Alcoholic Liver Disease Mortality

	All Population	By Gender		By Age Group			By Race Group		
	All	Male	Female	Young	Midlife	Old	White	Black	Other
<i>Panel A: Static DiD with case fixed effects (Greenstone and Moretti, 2003), Alcoholic Liver Disease</i>									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treated $\times$ After Treatment	-0.10 (0.51)	0.04 (0.77)	-0.22 (0.32)	0.02 (0.10)	-0.18 (0.71)	0.04 (0.90)	-0.20 (0.50)	0.28 (0.98)	-0.80 (0.82)
Mean	5.49	7.93	3.24	0.55	7.23	10.35	5.71	5.29	2.75
N	9258	9258	9258	9258	9258	9258	9258	9243	9258
<i>Panel B: Callaway–Sant’Anna ATT, Alcoholic Liver Disease</i>									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treated $\times$ After Treatment	0.01 (0.48)	0.29 (0.71)	-0.23 (0.32)	-0.09 (0.10)	0.23 (0.73)	0.05 (0.95)	-0.20 (0.51)	1.50* (0.78)	-0.61 (0.71)
Mean	5.49	7.93	3.24	0.55	7.23	10.35	5.71	5.29	2.75
N	9196	9196	9196	9196	9196	9196	9196	9181	9196

Notes: This table reports the estimated effect of winning a large manufacturing plant on alcoholic liver disease mortality (deaths per 100,000), for the full population and by gender, age, and race. Panel A reports static difference-in-differences estimates with county, case, and year fixed effects, following the Greenstone and Moretti (2003) winner-runner-up design; case fixed effects restrict comparisons to counties competing for the same plant. Panel B reports Callaway and Sant’Anna (2021) estimates robust to staggered adoption. All specifications are population-weighted, with standard errors clustered at the county level. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table A.7: Effect of Plant Openings on Local Labor Markets

	Log employment	Log establishments	Log total wages
<i>Panel A: Static DiD with case fixed effects (Greenstone and Moretti, 2003)</i>			
	(1)	(2)	(3)
Plant opening	0.097** (0.048)	0.079* (0.046)	0.094* (0.049)
N	8893	8893	8893
<i>Panel B: Callaway–Sant’Anna ATT</i>			
	(1)	(2)	(3)
Plant opening	0.098** (0.044)	0.083** (0.040)	0.098** (0.044)
N	8831	8831	8831

Notes: This table reports the estimated effect of winning a large manufacturing plant on local labor-market outcomes: log employment, log establishments, and log total wages (QCEW). Panel A reports static difference-in-differences estimates with county, case, and year fixed effects, following the Greenstone and Moretti (2003) winner-runner-up design; case fixed effects restrict comparisons to counties competing for the same plant. Panel B reports Callaway and Sant’Anna (2021) estimates robust to staggered adoption. All specifications are population-weighted, with standard errors clustered at the county level. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table A.8: Effect of Plant Openings on Overdose Mortality, Controlling for State Opioid Legislation

	All Population	By Gender		By Age Group			By Race Group		
	All	Male	Female	Young	Midlife	Old	White	Black	Other
<i>Static DiD with case fixed effects (Greenstone and Moretti, 2003), opioid-law $\times$ year, Overdose</i>									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Plant opening	-2.36** (1.13)	-3.62** (1.64)	-1.28* (0.72)	-1.23 (0.98)	-4.60** (2.07)	-1.58* (0.83)	-2.92** (1.36)	-2.12* (1.18)	-0.49 (0.68)
Mean	11.15	15.75	6.86	9.51	16.51	7.27	11.97	12.15	3.15
N	9258	9258	9258	9258	9258	9258	9258	9243	9258

Notes: This table reports the estimated effect of winning a large manufacturing plant on overdose mortality (deaths per 100,000), controlling for state opioid-control legislation. Following Meara et al. (2016) and Pierce and Schott (2020), state legislative stringency is measured as the number of opioid-related law categories in effect. Estimates are from the static difference-in-differences specification with county, case, and year fixed effects, interacting the opioid-law measure with year dummies. All specifications are population-weighted, with standard errors clustered at the county level. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table A.9: Robustness to Regional Timing of the Opioid Epidemic

	All Population	By Gender		By Age Group			By Race Group		
	All	Male	Female	Young	Midlife	Old	White	Black	Other
<i>Panel A: Static DiD with case fixed effects (Greenstone and Moretti, 2003), Overdose</i>									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Plant opening	-2.491** (1.197)	-3.801** (1.753)	-1.358* (0.752)	-1.349 (1.026)	-4.820** (2.227)	-1.723* (0.880)	-2.982** (1.447)	-2.642** (1.255)	-0.591 (0.699)
Mean	11.15	15.75	6.86	9.51	16.51	7.27	11.97	12.15	3.15
N	9258	9258	9258	9258	9258	9258	9258	9243	9258
<i>Panel B: + Region $\times$ Year FE, Overdose</i>									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Plant opening	-2.316** (1.121)	-3.356** (1.612)	-1.438* (0.730)	-1.338 (1.025)	-4.297** (2.020)	-1.562** (0.788)	-2.976** (1.328)	-1.790 (1.135)	-0.651 (0.660)
Mean	11.15	15.75	6.86	9.51	16.51	7.27	11.97	12.15	3.15
N	9258	9258	9258	9258	9258	9258	9258	9243	9258
<i>Panel C: + State $\times$ Year FE, Overdose</i>									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Plant opening	-1.531 (1.244)	-2.401 (1.867)	-0.818 (0.731)	-0.334 (0.839)	-3.387 (2.408)	-1.488 (1.078)	-2.254 (1.541)	-1.607 (1.367)	0.013 (0.342)
Mean	11.11	15.70	6.84	9.45	16.43	7.30	11.94	12.13	3.03
N	8858	8858	8858	8858	8858	8858	8858	8843	8858

Notes: This table reports the estimated effect of winning a large manufacturing plant on overdose mortality (deaths per 100,000), for the full population and by gender, age, and race. Panel A reports static difference-in-differences estimates with county, case, and year fixed effects, following the Greenstone and Moretti (2003) winner-runner-up design. Panel B adds region-by-year fixed effects and Panel C adds state-by-year fixed effects, absorbing differential regional and state-level timing of the opioid epidemic. All specifications are population-weighted, with standard errors clustered at the county level. The effect is robust to region-by-year fixed effects; under the most demanding state-by-year specification the point estimate is similar but less precisely estimated, reflecting the substantial identifying variation absorbed. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

Table A.10: Opioid Supply Balance: Winner vs. Loser Counties

	Total (1)	Oxycodone (2)	Hydrocodone (3)	Fentanyl (4)	Morphine (5)	Codeine (6)
Winner	1.33 (35.58)	2.64 (20.66)	3.96 (7.01)	-4.05 (6.20)	-1.20 (4.78)	-0.03 (0.71)
Mean supply (million MME)	205	98	37	39	26	4
Counties	162	162	162	162	162	162

Notes: This table reports pooled winner-vs-loser differences in opioid supply (in morphine milligram equivalents, MME), for total supply and by drug, over 1997-2017. Each column is a separate regression of county-year supply on a winner indicator with year fixed effects; standard errors are clustered at the county level. The coefficient on *Winner* is the average winner-minus-loser difference in supply. Small and statistically insignificant coefficients indicate no systematic difference in opioid availability between winner and loser counties, supporting the identifying assumption that the estimated mortality effects are not driven by differential drug supply. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

B Data Appendix

The International Classification of Diseases (ICD), developed by the World Health Organization (WHO), is the global standard for coding and classifying causes of death. Because this analysis spans 1970-2022, it draws on three ICD revisions:

1. **ICD-8** (8th Revision, 1965): a widely adopted revision providing three- to four-digit numeric codes; used to code U.S. mortality from 1968 to 1978.
2. **ICD-9** (9th Revision, 1977): expanded ICD-8 with up to five-digit numeric codes, adding categories and subdivisions to accommodate growing medical knowledge; used from 1979 to 1998.
3. **ICD-10** (10th Revision, 1990): introduced alphanumeric codes for greater specificity and to reflect advances in medical science; used from 1999 onward.

Rather than working directly from raw ICD codes, whose definitions and boundaries shift across revisions, I identify the three causes of death (drug overdose, suicide, and alcoholic liver disease) using the NCHS cause-of-death *recodes*, which map the underlying ICD codes into harmonized categories within each revision. This approach follows [Pierce and Schott \(2020\)](#). Specifically, I use the 281-cause recode for ICD-8 (1970-1978), the 282-cause recode for ICD-9 (1979-1998), and the 358-cause recode for ICD-10 (1999-2022). Table B.1 lists the recode categories assigned to each cause.

I depart from [Pierce and Schott \(2020\)](#) in two respects, both affecting the overdose category: I additionally include recode category 35300 in the ICD-9 recode and category 443 in the ICD-10 recode. For ICD-8, drug-overdose deaths of undetermined intent are captured through the underlying ICD-8 codes E980.0-E980.5, as these are not separately identified in the 281-cause recode.

Table B.1: Cause-of-Death Recode Categories

	ICD-8, 1970-78 (281-cause recode)	ICD-9, 1979-98 (282-cause recode)	ICD-10, 1999-2022 (358-cause recode)
Drug overdose	30600; E980.0-E980.5	31600-31900; 35300	419-422; 443
Suicide	32600-33000	33700-34400	424-431
Alcoholic liver disease	24200	24200	298

Notes: Entries are NCHS cause-of-death recode categories (and, for ICD-8 overdose of undetermined intent, the underlying ICD-8 codes). Cause definitions follow [Pierce and Schott \(2020\)](#), with the exception of categories 35300 (ICD-9) and 443 (ICD-10), which I add to the overdose group.